

80386 Register Organization -- MPCA Module 5

Definition: The 80386 Register Set consists of thirty-two 32-bit and 16-bit registers categorized into General-Purpose Registers, Segment Registers, Status and Instruction Registers (EFLAGS & EIP), Control Registers, and Debug/Test Registers.

1. Detailed Technical Explanation

1. General-Purpose Registers (32-bit Extended)

All 16-bit registers of 8086 were extended to 32 bits with the E prefix:

31	16	15	8	7	0	
+	+	+	+	+	+	
			AH		AL	EAX (Accumulator)
+	+	+	+	+	+	
			BH		BL	EBX (Base Register)
+	+	+	+	+	+	
			CH		CL	ECX (Count Register)
+	+	+	+	+	+	
			DH		DL	EDX (Data Register)
+	+	+	+	+	+	
				SI		ESI (Source Index)
+	+	+	+	+	+	
				DI		EDI (Destination Index)
+	+	+	+	+	+	
				BP		EBP (Base Pointer)
+	+	+	+	+	+	
				SP		ESP (Stack Pointer)
+	+	+	+	+	+	

2. Segment Registers (16-bit)

The 80386 has 6 Segment Selectors (added FS and GS for additional data segment access):

* CS
Segme

3. EFLAGS Register (32-bit Extended Flags)

Extends the 8086 flags with critical operating system mode control bits:

31	18	17	16	14	13-12	11	10	9	8	7	6	4	2	0
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+														
Reserved (0) AC VM RF 0 IOPL OF DF IF TF SF ZF AF PF CF														
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+														

* VM (Virtual 8086 Mode, Bit 17): When set (1), processor switches to Virtual 8086 mode.

* RF

4. Control Registers (CR0 - CR3)

* CR0 (Machine Status Word Extended):

* PE

2. Quick Recall Flow

80386 Registers -> 8 GPRs (EAX-ESP) -> 6 Segments (CS, DS, SS, ES, FS, GS) -> EFLAGS (VM, RF, IOPL, NT) -> Control (CR0: PE/PG, CR3: Page Directory)